

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

1. PHISHING

- Meaning: Fake messages, emails, or websites are used to steal passwords or personal information.
- Example: An attacker sends a fake WhatsApp or Instagram login link and tricks the user into entering their password.

2. DDoS (Distributed Denial-of-Service)

- Meaning: An attacker sends a huge amount of traffic to a service to make it slow or unavailable.
- Example: An attacker floods an online payment service such as Paytm with traffic, preventing legitimate users from accessing it.

3. MAN-IN-THE-MIDDLE (MITM)

- Meaning: An attacker secretly intercepts communication between two devices or users.
- Example: An attacker on unsecured public Wi-Fi intercepts a user's communication with an online service.

4. MALWARE

- Meaning: Malicious software that can damage a device or steal information.
- Example: A fake WhatsApp or Instagram update downloaded from an unsafe website installs malware on the user's phone.

5. PASSWORD ATTACKS

- Meaning: Attackers try to guess, steal, or reuse passwords to access accounts.
- Example: Attackers use leaked or reused passwords to attempt unauthorized access to an Instagram or Paytm account.

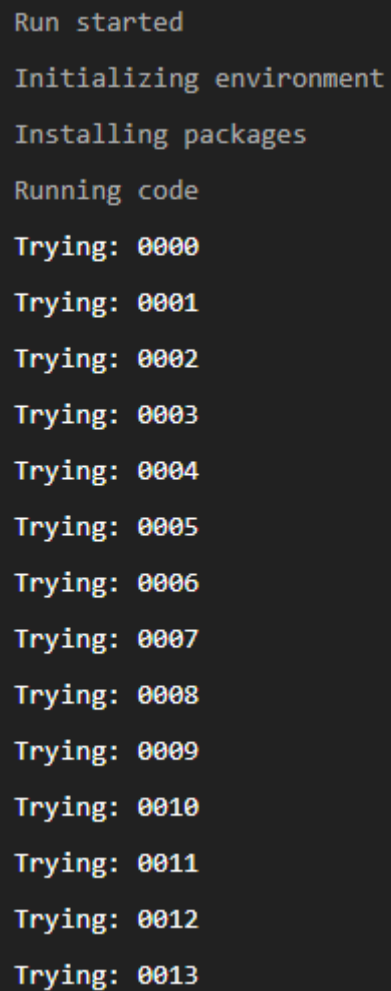
2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

```
# Brute-force PIN simulation
preset_pin = "5729"

for number in range(10000):
    guess = f"{number:04d}"

    print("Trying:", guess)

    if guess == preset_pin:
        print("PIN found:", guess)
        break
```



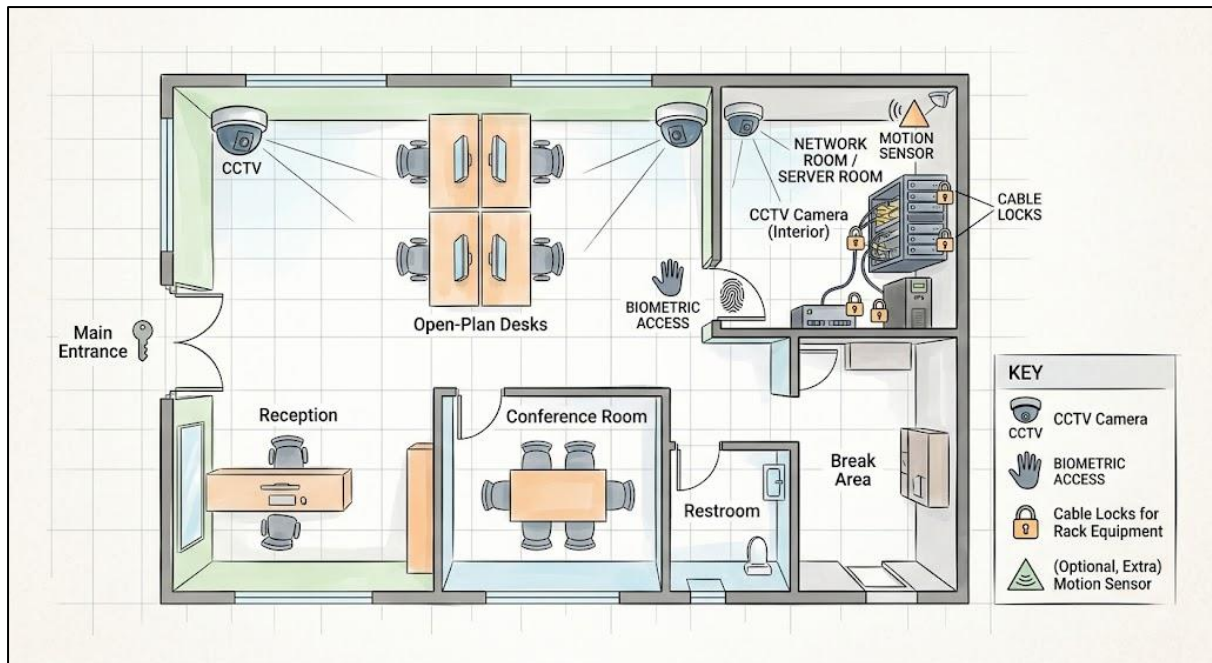
```
Run started
Initializing environment
Installing packages
Running code
Trying: 0000
Trying: 0001
Trying: 0002
Trying: 0003
Trying: 0004
Trying: 0005
Trying: 0006
Trying: 0007
Trying: 0008
Trying: 0009
Trying: 0010
Trying: 0011
Trying: 0012
Trying: 0013
```

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk

Physical Security Risk	Example for a Server Room	Mitigation
1. Unauthorized Access	An unauthorized person enters the server room and accesses servers or networking equipment.	Use biometric access control, security cameras, and access logs.
2. Fire	An electrical fault or overheating equipment causes a fire and damages servers.	Install smoke/fire detection and automatic fire-suppression systems.
3. Power Failure	A power outage causes servers hosting the app to shut down.	Use UPS systems and backup generators to maintain power.
4. Water Leakage/Flooding	A leaking pipe or cooling-system failure damages servers and cables.	Install water-leak sensors and keep critical equipment elevated above floor level.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

1. **Biometric Access:** Limits entry to the network room to authorized personnel only.
2. **CCTV Monitoring:** Provides continuous video surveillance of the critical server equipment.
3. **Cable Locks:** physically secure the network hardware and prevent theft or tampering.
4. **Motion Sensor:** Detects unauthorized activity in the server area outside of regular hours.



5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

No.	Action	How it applies to a college computer lab
1	Use strong passwords	Change default passwords on routers, switches, Wi-Fi, and lab administrator accounts.
2	Enable firewall protection	Configure firewalls on lab PCs and network devices to block unauthorized traffic.
3	Keep software updated	Regularly update Windows, antivirus, browsers, and other software to fix security vulnerabilities.
4	Secure Wi-Fi access	Use WPA2/WPA3, a strong Wi-Fi password, and separate student and administrator networks where possible.
5	Restrict physical access	Keep switches, routers, and servers in a locked network cabinet or restricted room.
6	Monitor for suspicious activity	Check network logs, antivirus alerts, and unusual login activity for signs of attacks or malware.